

CHUTE-DES-PASSES, QC, Canada

WMO: 715220

Lat: 49.840N

Lon: 71.168W

Elev: 1306

StdP: 14.02

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.3	-21.4	-33.1	0.9	-25.9	-28.0	1.2	-21.0	21.0	1.8	19.0	3.1	4.4	320	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	80.7	64.8	77.4	63.0	74.2	61.9	68.7	75.0	66.7	72.6	64.7	70.5	6.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	103.6	71.0	64.7	95.9	69.1	62.6	89.0	67.2	33.7	74.9	32.1	72.7	30.5	70.6	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	17.7	15.3	13.3	DB	-32.4	86.5	5.1	3.1	-36.1	88.7	-39.0	90.5	-41.9	92.2	-45.6	94.5
(5)				WB	-33.2	71.8	5.5	2.0	-37.2	73.2	-40.4	74.4	-43.4	75.5	-47.4	76.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.9	1.1	4.6	16.8	31.8	46.5	57.0	61.4	59.4	51.3	38.8	25.6	10.4	(6)
(7)		DBStd	23.09	13.46	12.95	12.96	9.29	8.12	7.30	5.22	5.58	7.82	7.47	9.24	12.33	(7)
(8)		HDD50	6946	1516	1272	1031	546	167	19	0	3	76	356	731	1229	(8)
(9)		HDD65	11426	1981	1692	1496	995	576	256	137	188	418	812	1181	1694	(9)
(10)		CDD50	1059	0	0	0	1	58	229	353	295	114	9	0	0	(10)
(11)		CDD65	64	0	0	0	0	2	16	25	16	5	0	0	0	(11)
(12)		CDH74	704	0	0	0	0	54	226	249	140	35	0	0	0	(12)
(13)		CDH80	109	0	0	0	0	10	45	35	16	3	0	0	0	(13)
(14)	Wind	WSAvg	5.5	5.4	5.8	6.3	6.7	6.1	5.5	4.9	4.7	4.9	5.5	5.5	5.1	(14)
(15)	Precipitation	PrecAvg	42.3	2.6	2.2	2.3	3.0	3.4	4.1	5.0	4.4	4.4	4.1	3.4	3.2	(15)
(16)		PrecMax	48.5	5.2	4.1	4.0	4.8	5.9	6.5	11.5	9.6	8.9	6.5	5.0	5.0	(16)
(17)		PrecMin	36.6	0.8	1.0	1.1	1.1	1.3	2.2	2.3	2.2	2.6	1.6	1.2	1.6	(17)
(18)		PrecStd	3.0	0.9	0.7	0.8	1.0	1.1	1.1	1.7	1.5	1.2	1.2	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.2	38.3	51.4	60.3	80.1	85.5	84.0	82.4	78.3	66.0	51.3	36.9	(19)
(20)			MCWB	38.0	35.5	42.2	45.8	58.9	66.0	66.8	66.6	67.5	57.6	49.1	36.1	(20)
(21)		2%	DB	32.8	33.5	42.7	54.9	73.3	80.0	80.0	78.2	72.8	59.0	45.2	33.8	(21)
(22)			MCWB	32.5	31.5	36.0	43.3	56.0	62.6	64.5	65.7	64.6	53.1	43.4	33.1	(22)
(23)		5%	DB	25.8	29.2	38.2	49.6	67.6	76.1	76.6	74.3	68.3	54.5	40.7	31.4	(23)
(24)			MCWB	25.0	27.6	33.6	40.6	53.2	60.4	63.3	63.7	61.1	50.5	39.0	30.9	(24)
(25)		10%	DB	21.2	25.0	34.7	45.0	62.2	71.7	73.2	70.7	64.4	50.5	37.3	27.7	(25)
(26)			MCWB	20.3	23.4	31.6	37.4	51.2	58.9	62.2	61.9	59.3	47.5	36.1	27.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.0	36.6	43.2	49.2	64.2	70.3	71.0	70.9	70.2	59.3	49.6	36.3	(27)
(28)			MDCB	38.1	37.0	46.1	54.9	71.0	79.2	78.0	76.9	76.1	64.3	50.7	36.7	(28)
(29)		2%	WB	32.5	31.5	37.4	44.4	58.7	67.0	68.2	68.0	66.4	54.8	44.1	33.2	(29)
(30)			MDCB	32.8	33.3	40.2	52.4	66.8	73.5	74.2	74.4	70.7	58.0	44.9	33.6	(30)
(31)		5%	WB	25.1	27.3	34.5	41.6	55.4	64.2	66.1	66.0	63.2	51.3	39.3	30.8	(31)
(32)			MDCB	25.6	29.0	37.5	48.1	64.3	71.0	72.2	71.9	67.3	53.4	40.3	31.4	(32)
(33)		10%	WB	20.3	23.3	31.9	38.8	52.4	61.2	64.2	63.8	60.2	47.8	36.0	27.1	(33)
(34)			MDCB	21.1	24.7	34.6	43.8	60.8	68.5	70.5	68.6	63.4	50.1	37.1	27.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.4	20.2	20.8	18.5	20.9	21.4	19.4	18.7	18.0	12.9	11.1	14.6	(35)
(36)			MCDBR	19.5	17.1	21.0	25.1	30.9	28.3	25.4	24.4	23.8	20.6	14.9	14.0	(36)
(37)			MCWBR	19.5	16.1	15.2	15.5	16.5	14.0	12.4	13.1	15.2	15.1	13.5	14.0	(37)
(38)		5% WB	MCDBR	19.7	17.4	17.7	21.8	25.3	22.0	19.4	19.9	20.4	17.9	13.6	14.0	(38)
(39)			MCWBR	19.9	16.3	13.8	15.1	15.4	13.7	11.3	12.2	14.4	14.4	13.1	14.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.248	0.263	0.290	0.338	0.340	0.371	0.380	0.366	0.352	0.312	0.289	0.245	(40)	
(41)		taud	2.294	2.267	2.237	2.256	2.446	2.398	2.391	2.435	2.450	2.547	2.452	2.347	(41)	
(42)		Ebn at Noon	258	282	292	287	291	281	277	276	268	263	241	243	(42)	
(43)		Edh at Noon	21	28	35	39	34	36	36	33	29	22	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	377	714	1119	1467	1676	1743	1591	1312	989	567	298	226	(44)	
(45)		RadStd	54	78	110	118	144	135	142	78	75	57	28	27	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page